

Notes: Jiang and Zhang (JFE, 2025)

Collateral Value Uncertainty and Mortgage Credit Provision

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1 Big picture

A residential mortgage is a debt contract collateralized by a house. Unlike liquid financial collateral (stocks/bonds), a house has no continuous “live price,” so lenders rely on appraisals. This creates a special tension:

- **Collateral recovery channel (standard).** If collateral values are more uncertain (and/or more illiquid), expected recovery in foreclosure is lower (or more dispersed). Competitive lenders then charge higher interest rates for the same leverage.
- **Appraisal risk channel (regulatory + measurement).** Mortgage LTV constraints (imposed by regulators and securitizers) are evaluated using the *minimum of transaction price and appraised value*. If appraisals are noisy and sometimes low, then even a willing borrower can be forced into a higher down payment. Some deals fail; many borrowers choose lower leverage ex ante to hedge appraisal risk.

The main empirical challenge is identification: worse outcomes in high-dispersion areas could reflect worse borrowers, not worse collateral. The paper’s strategy is to (i) measure dispersion at the property level using hedonic prediction errors, then (ii) instrument for dispersion with a “house non-standardization” measure that varies within county-year and is designed to be orthogonal to borrower credit quality.

2 Measurement of collateral value uncertainty: the price dispersion estimator

2.1 Hedonic price equation (mean of the price distribution)

Let p_{it} be the transaction price of property i sold at time t (log price in the paper). Let k index county and $x_{i,t}$ denote observed house characteristics (age, size, bedrooms, bathrooms, geo-coordinates, etc.). The paper estimates the conditional mean of prices via:

$$p_{it} = \eta_{kt} + f_k(x_{i,t}) + \varepsilon_{it}, \quad (1)$$

where η_{kt} is a county-by-month fixed effect, and $f_k(\cdot)$ is a flexible function (a generalized additive model) that maps characteristics into price.

2.2 Heteroskedasticity equation (variance of the price distribution)

The key idea is that *houses that are hard to value* will have large idiosyncratic pricing errors. So the paper models the conditional variance of ε_{it} by regressing the squared residuals on characteristics and time:

$$\widehat{\varepsilon}_{it}^2 = g_k(x_{i,t}) + \xi_{it}, \quad (2)$$

where $g_k(\cdot)$ is another flexible function.

2.3 Price dispersion as predicted idiosyncratic volatility

The *estimated price dispersion* for house i at time t is defined as the predicted standard deviation of the idiosyncratic component:

$$\hat{\sigma}_{it} \equiv \sqrt{\hat{g}_k(x_{i,t})}. \quad (3)$$

Interpretation. $\hat{\sigma}_{it}$ is a *house-level* measure of how noisy prices are for similar houses in the local market at that time. Economically, it is the “idiosyncratic” part of value uncertainty that matters for collateral, because most comparisons in the paper are within region-year cells.

Aggregation. Because of data-linking constraints across datasets, the paper aggregates house-level dispersion to the 5-digit zipcode-year level and assigns that zipcode dispersion to mortgages and applications in that location-year.

3 Stylized facts: what dispersion looks like and why appraisals matter

3.1 Figure 1: persistence and demographics

Fig. 1(a) shows zipcode dispersion in 2020 plotted against zipcode dispersion in 2010. The strong positive relation implies that dispersion is persistent over time: cross-sectional differences reflect slow-moving local housing-stock characteristics rather than short-lived shocks.

Fig. 1(b) compares dispersion across demographics (black-dominant vs non-black-dominant zipcodes, and low-income vs high-income, conditioning appropriately). The bars show that dispersion is higher in disadvantaged areas. This matters because any dispersion-induced credit tightening has distributional consequences.

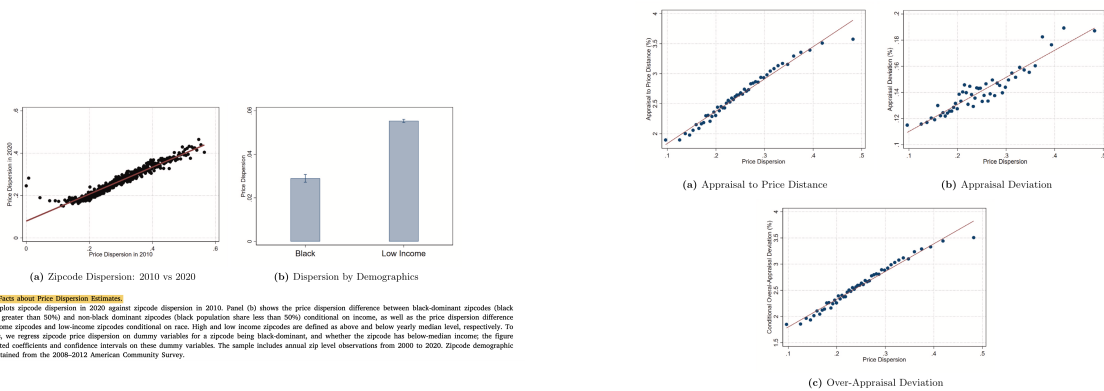


Fig. 1. Stylized Facts about Price Dispersion Estimates.
Notes: Panel (a) plots zipcode dispersion in 2020 against zipcode dispersion in 2010. Panel (b) shows the price dispersion difference between black-dominant zipcodes (black population share greater than 50%) and non-black dominant zipcodes (black population share less than 50%) conditional on income, as well as the price dispersion difference between high-income zipcodes and low-income zipcodes conditional on race. High and low income zipcodes are defined as above and below yearly median level, respectively. To obtain the values, we regress zipcode price dispersion on dummy variables for a zipcode being black-dominant, and whether the zipcode has below-median income; the figure shows the estimated coefficients and confidence intervals on these dummy variables. The sample includes annual zip level observations from 2000 to 2020. Zipcode demographic information is obtained from the 2000–2012 American Community Survey.

Fig. 2. Price Dispersion and Appraisals.
Notes: Panel (a) of this figure shows a binned scatter plot, where the y-variable is Appraisal to Price distance, defined as $\frac{p_i - a_i}{p_i}$. In panel (b), the y-variable is the appraisal deviation, defined as the product of the appraisal gap, and a dummy for under-appraisal, $\frac{p_i - a_i}{p_i} \cdot \mathbb{1}_{\{a_i < p_i\}}$. In panel (c), the y-variable is the average over-appraisal percentage conditional on over-appraisal. In all panels, the x-variable is zipcode price dispersion. We divide all loans into 50 buckets based on zipcode house price dispersion. The sample includes loan level observations from 2000 to 2020. Appraisal values are obtained from the CoreLogic LIMA data.

3.2 Figure 2: dispersion and appraisal noise

The appraisal system is central because LTV constraints depend on appraisals, not only on transaction prices.

Fig. 2(a): Appraisal-to-price distance increases with dispersion. This says appraisals become more error-prone when collateral is harder to value.

Fig. 2(b): “Appraisal deviation” (capturing the probability and size of *under*-appraisals) increases with dispersion. Under-appraisals are exactly what bind regulatory LTV constraints.

Fig. 2(c): Over-appraisal deviations also rise with dispersion, but the paper emphasizes that over-appraisals do not loosen borrowing much in practice because loan size is constrained by $\min\{P, A\}$ and by maximum LTV parameters; hence the *downside* appraisal tail is what matters for credit provision.

3.3 Table 1: data and magnitudes of dispersion

Table 1 reports summary statistics for three samples: property transactions (CoreLogic deeds/tax), loan-level mortgages (CoreLogic LLMA), and mortgage applications (HMDA).

The key takeaway is the scale of dispersion:

- Property-level dispersion has mean around 0.24 with standard deviation about 0.11.
- In mortgage applications, rejection rates average about 16%, while rejections explicitly due to collateral are rarer (around 2%), making them informative about the collateral channel.

Table 1
Summary statistics.
This table reports summary statistics for the three main datasets: the property sample from the CoreLogic Deed and Tax datasets, the loan sample from the CoreLogic LLMA dataset, and the mortgage application sample from the HMDA. The CoreLogic samples span the time period 2000 to 2020. The HMDA sample spans 2000 to 2017.

	N	Mean	Stdev	P25	Median	P75
Property Level Sample						
Loan to Price	29M	85.42	15.65	80.00	89.68	98.19
Price Dispersion	29M	0.24	0.11	0.17	0.23	0.30
Sale Price (Thousand)	29M	273.02	224.93	140.30	215.00	332.50
Mortgage Amount (Thousand)	29M	222.40	163.53	121.80	182.16	275.79
Building Age	29M	27.12	25.95	6.00	20.00	42.00
Square Footage	29M	1,961.57	2,982.11	1,363.00	1,774.00	2,365.00
Loan Level Sample						
Loan to Price	4.8M	85.48	14.98	80.00	90.00	98.19
Zip Price Dispersion	4.8M	0.25	0.08	0.19	0.24	0.29
Sale Price (Thousand)	4.8M	280.83	242.84	143.50	218.00	340.00
Appraised to Price Ratio	4.8M	1.03	0.19	1.00	1.00	1.02
Mortgage Amount (Thousand)	4.8M	227.66	170.25	124.00	185.18	283.00
FICO	4.8M	725.35	61.39	681.00	735.00	778.00
Debt-to-Income	4.8M	37.23	11.28	29.85	38.00	44.69
Mortgage Application Sample						
Rejection Rate	49M	15.86	36.53	0.00	0.00	0.00
Rejection due to Collateral Reasons	49M	1.95	13.83	0.00	0.00	0.00
Zip Price Dispersion	49M	0.26	0.08	0.20	0.25	0.31
Applicant Income (Thousand)	49M	102.35	193.47	47.00	72.00	114.00
Loan-to-Income	49M	242.18	6,896.19	135.83	227.78	316.51
County Credit Score	49M	667.19	22.16	650.30	666.18	684.14

Table 2
Determinants of house price dispersion.
This table presents the association between house price dispersion and house features (Panel A) and zipcode market condition (Panel B). All continuous variables are normalized by their standard deviations. We define a house as recently renovated if it has been renovated within 5 years before the transaction year. The sample includes house transactions from 2000 to 2020. Standard errors are clustered at county level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

Panel A: House Features					
	Estimated Price Dispersion				
	(1)	(2)	(3)	(4)	(5)
Building Age	0.04*** (0.002)	0.04*** (0.002)	0.04*** (0.002)		0.04*** (0.002)
Recent Renovation		-0.01*** (0.003)			-0.01*** (0.003)
Benchmark: Square-Footage < 1281					
[1282,1601]			-0.03*** (0.002)	-0.03*** (0.002)	-0.02*** (0.002)
[1602,1970]			-0.03*** (0.003)	-0.03*** (0.003)	-0.02*** (0.003)
[1971,2544]			-0.02*** (0.004)	-0.02*** (0.004)	-0.00 (0.003)
> 2544			0.01** (0.005)	0.01** (0.004)	0.03*** (0.004)
Benchmark: Bedrooms < 4					
=4				-0.01*** (0.001)	-0.01*** (0.001)
> 4				0.01*** (0.002)	0.01*** (0.001)
Log House Price	-0.48*** (0.028)	-0.48*** (0.028)	-0.51*** (0.025)	-0.51*** (0.025)	-0.37*** (0.021)
Log House Price Squared	0.50*** (0.029)	0.50*** (0.029)	0.51*** (0.025)	0.51*** (0.025)	0.38*** (0.022)
County-Year FE	✓	✓	✓	✓	✓
R2	0.33	0.33	0.26	0.27	0.35
Observations	29M	29M	29M	29M	29M
Panel B: Zipcode Market Conditions					
	Zipcode Price Dispersion				
	(1)	(2)	(3)	(4)	
Gini Index	0.01*** (0.001)				0.01*** (0.001)
Population Density		-0.01*** (0.003)			-0.01*** (0.002)
Vacancy Share			0.03*** (0.002)		0.03*** (0.002)
Year FE	✓	✓	✓	✓	✓
R2	0.02	0.02	0.08		0.09
Observations	276,079	276,079	276,079		276,079

3.4 Table 2: determinants of house price dispersion (what makes collateral “hard to value”)

Table 2 connects the dispersion measure back to observable house and neighborhood features. All continuous regressors are standardized, so coefficients can be read as “s.d.-to-s.d.” associations.

Panel A: house features. Dispersion is higher for houses that are *harder to benchmark against close comparables*:

- **Older houses:** building age has a strong positive association with dispersion (about 0.04 per s.d.).
- **Renovation reduces uncertainty:** recent renovation is associated with lower dispersion (about -0.01), consistent with renovations making quality more observable.
- **Size and configuration non-linearities:** relative to small houses, mid-size square-footage bins have lower dispersion while very large houses have higher dispersion; similarly, very high bedroom counts are associated with higher dispersion.
- **Price level nonlinearity:** the coefficients on $\log(\text{price})$ and its square imply a U-shape: dispersion is higher for very cheap or very expensive houses, lower in the middle of the price distribution.

Panel B: zipcode market conditions. Dispersion is also higher when the local market is *more segmented or more illiquid*:

- **Higher inequality (Gini) $\Rightarrow$ higher dispersion:** consistent with more heterogeneity in housing quality and buyer demand.
- **Higher vacancy share $\Rightarrow$ higher dispersion:** consistent with thin markets and noisier price signals.
- **Higher population density $\Rightarrow$ lower dispersion:** consistent with more transactions and better comparables (more information).

Why Table 2 matters for the later IV design. The IV strategy uses *feature distance from the county-year mean* (“non-standardization”) to shift dispersion. Table 2 provides the economic justification: non-standard houses (extreme size/configuration/age) are systematically associated with larger pricing errors, hence higher dispersion.

4 Reduced-form evidence: dispersion and mortgage credit provision

The paper studies three main outcomes, each naturally linked to a point in the mortgage-origination pipeline (Fig. 3):

1. **Mortgage rejection rates** (especially collateral-related rejections) — whether borrowers can obtain a mortgage at all.
2. **Interest rates controlling for leverage** — the “cost menu” faced by borrowers, i.e., rates at a given loan-to-price (LTP).
3. **Loan-to-price ratios (LTP)** — the equilibrium leverage choice/outcome, shaped by menus and appraisal risk.

Table 3
Mortgage rejections and zip house price dispersion.
This table presents loan level regression results about mortgage rejections. The outcome variable in Panel A is an indicator that equals 100 if a loan is rejected and 0 otherwise. The outcome variable in Panel B is an indicator that equals 100 if a loan is rejected due to collateral reasons and 0 otherwise. In both panels, columns 1-3 report OLS results, and columns 4-6 report 2SLS results. The explanatory variable of interest is zipcode house price dispersion in columns 1-3 and is the predicted zipcode price dispersion in columns 4-6, all scaled by its standard deviation. IV construction is described in Section 3.2.1. IV first stage results are presented in Table I.2. Borrower/Loan controls include zipcode house price, credit score and the squared term, log income, loan type, and loan to income ratio and its square term. The sample includes loan level observations from 2000 to 2017. Standard errors are clustered at county level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	OLS			2SLS		
	(1) Full	(2) Securitized	(3) Portfolio	(4) Full	(5) Securitized	(6) Portfolio
Panel A: Rejection						
Zip Price Dispersion	1.40*** (0.093)	1.42*** (0.103)	0.81*** (0.116)	2.31*** (0.221)	2.30*** (0.234)	2.22*** (0.433)
Local Controls	✓	✓	✓	✓	✓	✓
County-Year FE	✓	✓	✓	✓	✓	✓
Lender-Year FE	✓	✓	✓	✓	✓	✓
Rejection Mean	15.9%	16.4%	16.2%	15.9%	16.4%	16.2%
R2	0.16	0.18	0.17	-	-	-
Observations	47M	34M	3.6M	47M	34M	3.6M
Underidentification t-stat				77.87	74.65	19.77
Underidentification p-value				0.00	0.00	0.00
Weak identification t-stat				30.75	29.81	13.74
Panel B: Rejection Due to Collateral						
Zip Price Dispersion	0.50*** (0.036)	0.54*** (0.038)	0.38*** (0.054)	0.79*** (0.065)	0.86*** (0.073)	0.73*** (0.095)
Local Controls	✓	✓	✓	✓	✓	✓
County-Year FE	✓	✓	✓	✓	✓	✓
Lender-Year FE	✓	✓	✓	✓	✓	✓
Rejection due to Collateral Mean	2.0%	2.0%	2.3%	2.0%	2.0%	2.3%
R2	0.05	0.05	0.09	-	-	-
Observations	47M	34M	3.6M	47M	34M	3.6M
Underidentification t-stat				77.87	74.65	19.77
Underidentification p-value				0.00	0.00	0.00
Weak identification t-stat				30.75	29.81	13.74

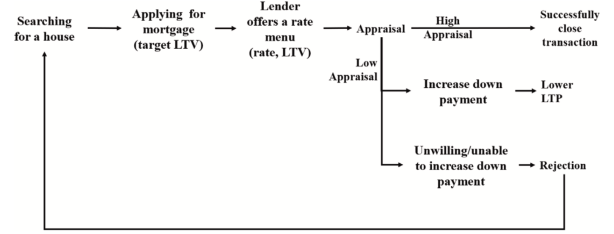


Fig. 3. Home Purchase - Mortgage Origination Diagram.

4.1 Figure 3: home purchase / mortgage origination diagram

Fig. 3 lays out the logic: a buyer searches for a house, applies for a mortgage choosing a target LTV/LTP, lenders offer a price menu (rate as a function of LTP and local dispersion), an appraisal occurs, and then either (i) a high appraisal allows the transaction to close at the planned terms, or (ii) a low appraisal forces a higher down payment, triggering either a lower realized LTP or a failure/rejection if the buyer cannot or will not make up the gap.

This diagram is important because the later structural model mirrors exactly these stages.

4.2 Mortgage rejection regressions (application-level)

At the application level, the paper estimates within county-year variation:

$$Reject_{ikt} = \beta ZipDispersion_{ikt} + X_{ikt}\Gamma + \mu_{kt} + \nu_{\ell t} + \epsilon_{ikt}, \quad (4)$$

where $Reject_{ikt}$ equals 100 if the application is rejected and 0 otherwise; $ZipDispersion_{ikt}$ is the average dispersion in the borrower’s zipcode in year t ; X_{ikt} are borrower and loan controls; μ_{kt} are county-year fixed effects; and $\nu_{\ell t}$ are lender-year fixed effects.

4.2.1 Table 3: dispersion raises rejections, especially collateral-related

Panel A (overall rejection). In OLS (cols. 1–3), a one-standard-deviation increase in zipcode dispersion raises rejection rates by about 0.8–1.4 percentage points (pp) depending on the loan type. In IV/2SLS (cols. 4–6), the effect roughly doubles, around 2.2–2.3 pp. Given a mean rejection rate around 16%, a 2.3pp increase is economically large (about a 14% relative increase).

Panel B (collateral-related rejection). Dispersion has an even clearer interpretation here because the outcome uses HMDA rejection reasons. In OLS, a one-s.d. increase in dispersion raises collateral-related rejection probability by roughly 0.38–0.54pp; in IV it rises to about 0.73–0.86pp. Relative to a mean of roughly 2%, this is a large proportional effect (roughly 35%–40%).

Interpretation. The collateral interpretation is strengthened by two patterns: (i) the effect is present for both securitized and portfolio loans, and (ii) it is strongest when the rejection reason is collateral-related. This pushes against the alternative story that dispersion is only proxying for borrower creditworthiness.

4.3 Interest rate “cost menu” regressions (loan-level)

To measure pricing conditional on leverage, the paper estimates:

$$Rate_{ikt} = \beta_1 ZipDispersion_{ikt} + \beta_2 LTP_{ikt} + X_{ikt}\Gamma + \mu_{kt} + \nu_m + \epsilon_{ikt}, \quad (5)$$

where $Rate_{ikt}$ is the interest rate (in basis points) on a mortgage collateralized by house i in county k , year t . The specification includes origination-month fixed effects ν_m and rich borrower/loan controls.

4.3.1 Figure 5: the entire menu shifts upward in high-dispersion zipcodes

Fig. 5 plots residualized interest rates against LTP separately for high- and low-dispersion zipcodes. The high-dispersion curve is above the low-dispersion curve at essentially every LTP, consistent with an upward shift in the cost menu.

4.3.2 Table 4: dispersion raises rates conditional on LTP

Table 4 reports OLS and IV versions of (5). A one-s.d. increase in dispersion raises interest rates by about 0.9 bps in OLS (full sample) and about 1.4 bps in IV. Effects are modest in level terms compared to rejection effects, but the menu-shift is systematic and consistent across loan types.

Interpretation. In the paper’s framework, the rate effect is the signature of the *collateral recovery* channel: higher dispersion implies worse expected recovery (or more downside risk) for a given LTP, so lenders price the loan more expensively.

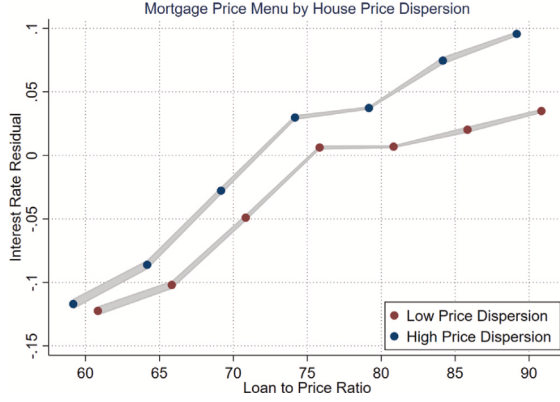


Fig. 5. Property Level Mortgage Menu by House Price Dispersion.

Notes: This figure shows the mortgage price menu (rate-LTP pair) by zip-level house price dispersion. The y-values are interest rate residuals from a regression of mortgage rates on borrower FICO, FICO-squared, DTI, DTI-squared conforming or jumbo indicator, and origination month fixed effects. The dots represent the average mortgage rate in each LTP bucket. The shaded area indicates a 95% confidence interval. The sample includes loan level observations of conventional loans in the Corelogic LLMA from 2000 to 2020.

Table 4

Price dispersion and cost menu.

This table presents loan-level regression results about the “cost menu”, that is, interest rates controlling for LTPs. The outcome variable is loan-level interest rate, in bps. Columns 1–3 present OLS results, and columns 4–6 present 2SLS results. Columns 1 and 4 use the full sample. Columns 2 and 5 use securitized conventional loans (i.e., non-FHA loans that are securitized). Columns 3 and 6 use portfolio conventional loans (i.e., non-FHA loans that are held on lenders’ balance sheets). The explanatory variable of interest is zipcode house price dispersion in columns 1–3 and is the predicted zipcode price dispersion in columns 4–6, all scaled by its standard deviation. IV construction is described in Section 3.2.1. IV first stage results are presented in Table I.2. Borrower and loan controls include log house price, FICO score, FICO squared, LTV, LTV squared, DTI, DTI-squared, and loan type. The sample includes loan level observations from 2000 to 2020. Standard errors are clustered at county level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	OLS			2SLS		
	(1) Full	(2) Securitized	(3) Portfolio	(4) Full	(5) Securitized	(6) Portfolio
Zip Price Dispersion	0.89*** (0.123)	1.31*** (0.109)	1.26*** (0.372)	1.42*** (0.306)	1.68*** (0.331)	3.03*** (0.683)
LTP	0.69*** (0.090)	0.58*** (0.039)	1.89*** (0.186)	0.69*** (0.090)	0.58*** (0.039)	1.88*** (0.184)
Borrower and Loan Controls	✓	✓	✓	✓	✓	✓
Origination Month FE	✓	✓	✓	✓	✓	✓
County-Year FE	✓	✓	✓	✓	✓	✓
R2	0.87	0.88	0.87	–	–	–
Observations	4.8M	2.3M	1.1M	4.8M	2.3M	1.1M
Underidentification t-stat				68.08	71.05	60.10
Underidentification p-value				0.00	0.00	0.00
Weak identification t-stat				23.17	20.77	18.35

4.4 Loan-to-price regressions (property-level)

The leverage outcome is measured as loan-to-price ratio (LTP), and the core property-level regression is:

$$LTP_{ikt} = \delta Dispersion_{ikt} + X_{ikt}\Gamma + \mu_{kt} + \nu_d + \epsilon_{ikt}, \quad (6)$$

where $Dispersion_{ikt}$ is the estimated dispersion of the underlying house, and ν_d are transaction-date fixed effects (and in richer specifications lender-year fixed effects).

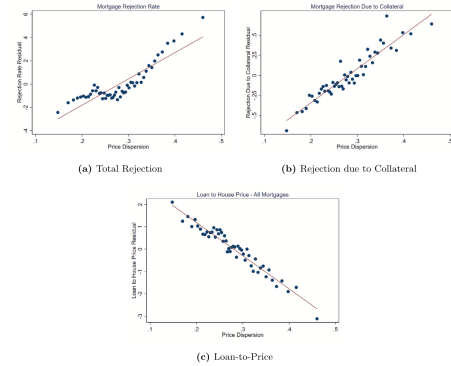


Fig. 4. County Level House Price Dispersion and Credit Access

Notes: This figure shows the correlation between county level house price dispersion and various credit access outcomes. The y-variable in panels (a) and (b) are residualized county-level mortgage rejection rates and county-level rates of rejection due to collateral-related reasons, respectively, obtained from the HMDA data. We take the residuals of regressions of county-level rejection rates, or rates of rejection due to collateral-related reasons on county average log house price, credit score, and year fixed effects. In panel (c) the y-variable is the average county-level LTP residual from a regression of county average LTP on county level house price. The y-variable is obtained from the Corelogic LLMA data, and values are in percentage points. The underlying samples in all three panels include annual county observations from 2000 to 2017.

Table 5

Property-level house price dispersion and LTP.

This table presents property-level regression results on the relationship between price dispersion and mortgage LTPs. Columns 1–3 present OLS results. Columns 4–6 present IV results. In all columns, the outcome variable is the loan level loan-to-sale-price ratio. The explanatory variable of interest in columns 1–3 is property-level house price dispersion, scaled by its standard deviation, and is the predicted price dispersion in columns 4–6. IV construction is described in Section 3.2.1. IV first stage results are presented in Table I.2. Controls include the transaction price of the property, mortgage type, mortgage term, and resale indicator. The sample includes property transaction level observations from 2000 to 2020. Standard errors are clustered at county level. ***, **, * represent 1%, 5%, and 10% significance, respectively.

	OLS			2SLS		
	(1)	(2)	(3)	(4)	(5)	(6)
Price Dispersion	-0.43*** (0.042)	-0.21*** (0.036)	-0.22*** (0.033)	-1.25*** (0.118)	-1.30*** (0.108)	-1.30*** (0.104)
Controls	✓	✓	✓	✓	✓	✓
Transaction Date FE	✓	✓	✓	✓	✓	✓
County-Year FE	✓	✓	✓	✓	✓	✓
Lender-Year FE	✓	✓	✓	✓	✓	✓
R2	0.34	0.36	0.40	–	–	–
Observations	28M	28M	28M	28M	28M	28M
Underidentification test statistic				175.44	162.02	165.68
Underidentification test p-value				0.00	0.00	0.00
Weak identification test statistic				189.39	202.32	197.81

4.4.1 Table 5: dispersion lowers LTP, with larger IV estimates

Table 5 shows that a one-s.d. increase in house-level dispersion reduces LTP by about 0.2–0.4pp in OLS, but by about 1.25–1.30pp in IV/2SLS. The larger IV estimates are consistent with attenuation bias in OLS due to measurement error in dispersion.

Interpretation. The negative LTP effect combines both channels: (i) higher interest rates make leverage more expensive, and (ii) higher under-appraisal risk makes high target leverage less attractive because it raises the probability of needing to unexpectedly increase the down payment.

4.5 Figure 4: county-level correlations

Fig. 4 provides a macro view (county-year level): dispersion is positively associated with residualized rejection (overall and collateral-related) and negatively associated with residualized LTP. This matches the micro regressions and visually supports the pipeline in Fig. 3.

5 Identification: the “house non-standardization” instrument

5.1 Motivation

A concern is that dispersion could proxy for borrower quality: less creditworthy borrowers might buy houses that are harder to value (or live in neighborhoods with more heterogeneity). If so, rejection and higher rates could reflect borrower risk, not collateral uncertainty.

The paper addresses this with an instrument that aims to shift dispersion while holding borrower characteristics fixed.

5.2 Constructing instruments from feature distance

For each house, define the squared distance between a feature and the county-year average of that feature:

$$Z_i^m = (X_i^m - \bar{X}^m)^2, \quad m \in \{\text{age, sq.ft., bed, bath, geo}\}. \quad (7)$$

Because some features are not symmetrically distributed (so $(X - \bar{X})^2$ could co-move with the level of X), the paper purges the component linearly related to house features by running:

$$Z_i^m = \sum_k \alpha_k X_i^k + \mu_d + e_i^m, \quad (8)$$

where μ_d are transaction-date fixed effects. The instrument is the residual:

$$\tilde{Z}_i^m = e_i^m. \quad (9)$$

Economic meaning. $\tilde{Z}_i^m$ is a measure of how “unusual” house i is relative to other houses transacted in the same county-year, along feature m . A house that is unusual should be harder to value, increasing dispersion, but the unusualness is not meant to be directly related to the borrower’s creditworthiness.

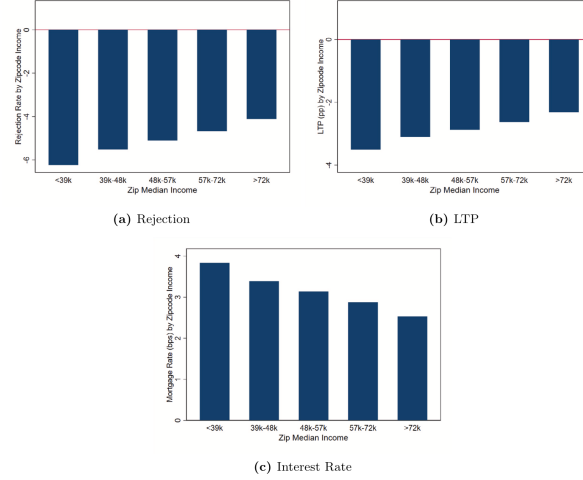


Fig. 6. Cross-Sectional Heterogeneity.

Notes: This figure presents how rejection rate, down payment, and mortgage rate would change if the house's price dispersion decreased to be equal to the bottom decile of price dispersion in our dataset. The x-axis are five equal-sized zipcode median income buckets, where zipcode median income is obtained from 2008-2012 American Community Survey. The y-variable is the impact on mortgage rejection rate in Panel (a), the impact on LTP in Panel (b), and the impact on interest rate in Panel (c). To obtain these values, we calculate the difference between a house's price dispersion and the bottom decile of price dispersion, and then multiply it respectively by the estimated effect of price dispersion on mortgage rejection (column 4 of Table 3), by the estimated effect on LTP (column 6 of Table 5), and by the estimated effect on mortgage rate (column 4 of Table 4). We find the average values in each zipcode income quintile.

5.3 2SLS structure

Let $Dispersion_{ikt}$ denote the endogenous dispersion measure (property or zipcode). Using the five instruments $\tilde{Z}_i^m$, the paper estimates:

$$\begin{aligned}
 \text{Stage 1: } Dispersion_{ikt} &= \beta_1 \tilde{Z}_i^{age} + \beta_2 \tilde{Z}_i^{sq.ft.} + \beta_3 \tilde{Z}_i^{bed} + \beta_4 \tilde{Z}_i^{bath} + \beta_5 \tilde{Z}_i^{geo} + Controls_{ikt} \Gamma_1 + \mu_{kt} + \nu_d + \epsilon_{ikt}, \\
 \text{Stage 2: } Y_{ikt} &= \gamma \widehat{Dispersion}_{ikt} + Controls_{ikt} \Gamma_2 + \eta_{kt} + \zeta_d + \xi_{ikt},
 \end{aligned} \tag{10}$$

where Y_{ikt} is one of the mortgage outcomes (rejection, rate, LTP).

5.4 Table 6: ex-post default as an exclusion test

Table 6 uses ex-post default within two years as an outcome (set to 100 for default). If dispersion were mostly borrower risk, it should predict default even in IV.

OLS: dispersion is positively related to default (coefficients around 0.15–0.21), consistent with selection. **IV/2SLS:** the dispersion coefficients become small and statistically insignificant (near zero). This supports the claim that the instrument shifts dispersion without shifting borrower credit risk.

Why this matters for the core results. Once the exclusion restriction is plausible, the larger IV effects in Tables 3–5 are naturally interpreted as correcting attenuation bias from measurement error in dispersion.

6 Distributional consequences: why low-income areas matter

Fig. 6 translates the reduced-form coefficients into a counterfactual thought experiment: for each transacted house, compute how rejection, LTP, and rates would change if its dispersion were reduced

to the bottom decile, then average within zipcode income buckets.

The bars imply that lowering dispersion would:

- reduce rejection rates more in low-income zipcodes than in high-income zipcodes,
- increase leverage (LTP) more in low-income zipcodes,
- lower interest rates more in low-income zipcodes.

This links the descriptive fact from Fig. 1(b) (dispersion is higher in disadvantaged areas) to an economically meaningful distributional mechanism.

7 Structural model: mapping dispersion into credit outcomes through two channels

The reduced-form evidence establishes that dispersion matters and is not (only) borrower quality. The structural model then provides a disciplined way to separate mechanisms.

7.1 Timeline and core state variables

The model follows Fig. 3:

1. The buyer chooses a target mortgage size L (equivalently, target LTV/LTP $l \equiv L/P$) for a house sold at price P .
2. Lenders offer an interest rate menu $r(L, \sigma)$ that depends on loan size and price dispersion σ .
3. An appraisal A is realized; regulatory and securitization rules impose that the LTV is computed using $\min\{P, A\}$ and capped by a maximum LTV parameter ϕ .
4. If the appraisal is too low, the buyer may (i) increase down payment and continue (possibly at a lower realized LTV), or (ii) renege (transaction fails), pay a cost, and search again.

7.2 Preferences (buyer)

The buyer has CRRA utility in period 1 and linear utility in period 2 (a tractability device):

$$U(c_1, c_2) = \frac{c_1^{1-\eta} - 1}{1-\eta} + \beta^T u'_2 c_2, \quad (11)$$

where η is risk aversion (inverse of intertemporal elasticity of substitution), β is the discount factor, T is the effective mortgage horizon, and u'_2 is a reduced-form marginal utility weight for period 2 consumption.

7.3 Loan size constraint from appraisals and regulation

Let the maximum LTV parameter be ϕ (e.g., $\phi = 0.8$). Because LTV is computed using $\min\{P, A\}$, the final loan size is:

$$L_{\text{final}} \leq \phi \min\{P, A\}. \quad (12)$$

Normalizing by price P :

$$l_{\text{final}} \equiv \frac{L_{\text{final}}}{P}, \quad l \equiv \frac{L}{P}, \quad a \equiv \frac{A}{P}. \quad (13)$$

Thus l is the target LTV/LTP, l_{final} is the realized LTV/LTP, and a is the appraisal-to-price ratio.

If $a < 1$, then the appraisal binds and:

$$\phi \min(P, A) = \phi P \min(1, a) = \phi a P, \quad (14)$$

so the buyer cannot borrow as much as planned.

7.4 Consumption under under-appraisal

The buyer planned a down payment $P(1 - l)$, but if $a < 1$, she must make an *additional* down payment of $P(l - \phi a)$ to reach the regulatory LTV cap. The paper models the cost of an unexpected increase in down payment with a penalty factor $\psi > 1$ applied to the additional amount:

$$c_1 = W_1 - P(1 - l) - \psi P \max\{0, l - \phi a\}, \quad (15)$$

$$c_2 = W_2 - (1 + r(l))^T P \left[l - \max\{0, l - \phi a\} \right]. \quad (16)$$

Here W_1, W_2 are exogenous incomes, and $r(l)$ denotes the interest rate implied by the menu at target leverage l .

Key mechanism in (15). When a is low, the required additional down payment is large, and because it is *unanticipated*, it is more painful than an equivalent planned down payment (captured by $\psi > 1$). This is the core of the *appraisal risk* channel: noisy appraisals create states where down payments jump unexpectedly.

7.5 Reneging (transaction failure)

If the appraisal is sufficiently low, the buyer may renege, paying a cost ζ (as a fraction of price) and restarting the search. The continuation value is:

$$-\beta^T u'_2 \zeta P + \mathbb{E}_a[V(a, l)], \quad (17)$$

where $V(a, l)$ is the value of choosing l when appraisal is a .

7.6 Interest rate menu (reduced form)

The paper assumes a reduced-form menu where rates rise with leverage and dispersion:

$$r(l, \sigma) = \bar{r} + \theta_l l + \theta_\sigma \sigma. \quad (18)$$

In calibration it is convenient to center the menu at $l = 0.8$ and at a benchmark dispersion $\bar{\sigma}$:

$$r(l, \sigma) = \bar{r} + \theta_l(l - 0.8) + \theta_\sigma(\sigma - \bar{\sigma}). \quad (24)$$

Interpretation. $\theta_\sigma > 0$ captures the *collateral recovery* channel in reduced form: more dispersion means lenders require a higher rate even at the same leverage. $\theta_l > 0$ captures standard leverage pricing.

7.7 Appraisal distribution with bunching at the transaction price

The paper models the empirical fact that appraisals bunch at sale prices and are asymmetrically distorted. Let the raw appraisal be:

$$A_{\text{raw}} \sim \mathcal{N}(P, \sigma_a),$$

and the realized appraisal A follows:

$$A = \begin{cases} A_{\text{raw}} + Pb, & A_{\text{raw}} < P(1 - b), \\ P, & P(1 - b) \leq A_{\text{raw}} < P, \\ A_{\text{raw}}, & P \leq A_{\text{raw}}. \end{cases} \quad (19)$$

Here $b \in (0, 1)$ is the upward bias intensity for low appraisals.

Interpretation. When raw appraisals fall moderately below P , appraisers report exactly P (bunching), eliminating small under-appraisals. When raw appraisals are far below P , appraisers push them up by Pb but still below P (partial bias). When appraisals exceed P , appraisers do not bias upward further because doing so does not increase borrowing (loan size is limited by $\min\{P, A\}$).

7.8 Cutoff structure: Theorem 1 and its intuition

Theorem 1 (Cutoff appraisal rule (paper's Theorem 1)). *For any parameters and any target loan size l , there exists a unique cutoff $\bar{a}(l)$ such that the buyer continues the purchase if $a \geq \bar{a}(l)$ and reneges if $a < \bar{a}(l)$. The cutoff satisfies:*

$$\omega(\bar{a}(l), l) = -\beta^T u'_2 \zeta P + \int_0^\infty \max\{\omega(a, l), \omega(\bar{a}(l), l)\} dF_a(a), \quad (20)$$

where $\omega(a, l)$ is the continuation value from continuing given a :

$$\omega(a, l) = u_1(W_1 - P(1 - l) - \psi P \max\{0, l - \phi a\}) + u'_2 \beta^T (1 + r(l))^T P \max\{0, l - \phi a\}. \quad (21)$$

Proof sketch (monotonicity $\Rightarrow$ unique cutoff). Fix l . Under (15)–(16), higher a weakly relaxes the binding constraint $\max\{0, l - \phi a\}$, raising c_1 and (weakly) raising c_2 . Because $u_1(\cdot)$ is strictly increasing and concave, and the period-2 term is linear in c_2 , $\omega(a, l)$ is continuous and strictly increasing in a on the region where the constraint binds. The reneging option yields a constant value (given l) up to the expected continuation term. Therefore the set of a for which continuing dominates reneging is an interval $[\bar{a}(l), \infty)$ with a unique boundary point.

7.9 Optimal target leverage choice

Given the cutoff rule, the buyer chooses l to maximize expected utility:

$$l^* = \arg \max_l \left(-\beta^T (1 + r(l))^T u'_2 P l + \int_0^\infty \max\{\omega(a, l), \omega(\bar{a}(l), l)\} dF_a(a) \right). \quad (22)$$

This captures the tradeoff: larger l smooths consumption when appraisals are high but raises interest costs and increases exposure to under-appraisal shocks.

8 Calibration and model fit

8.1 Externally calibrated parameters

Table 7 Panel A sets standard values: $\eta = 2$, $W_1 = \$60,000$, $P = \$200,000$, $\beta = 0.96$, $T = 7$ (approximate duration), and $\phi = 0.8$ (regulatory cap).

8.2 Estimating appraisal noise by dispersion buckets

A key object is the mapping from local dispersion σ to appraisal noise σ_a . The paper estimates σ_a using the conditional variance of appraisal ratios *on over-appraisal*, because over-appraisals are not distorted by (19):

$$\hat{\sigma}_{a,i} = \sqrt{\mathbb{E}[(a_i - 1)^2 \mid a_i > 1]}. \quad (23)$$

This is computed for each decile of σ .

8.3 Moment matching

The remaining parameters (search cost ζ , appraisal bias b , under-appraisal penalty ψ , and u'_2) are chosen to match:

- mortgage failure probabilities by σ -decile,
- appraisal deviation moments by σ -decile:

$$AppDev_i = p_i^{under} \mathbb{E}[1 - a_i \mid a_i < 1], \quad (25)$$

- the slope of the regression of average final leverage l_{final} on σ (“ l_{final} -to- σ coefficient”), using Table 5 as the empirical target.

8.4 Table 7: parameter estimates

Table 7 Panel B reports:

- $\zeta \approx 0.583$ (transaction failure is costly),
- $b \approx 0.086$ (a modest but precisely estimated upward bias for low appraisals),
- $\psi \approx 2.782$ (unexpected down payments are more painful than planned ones),
- $u'_2 \approx 0.00243$ (pins down borrowing demand via Euler-equation-like moments).

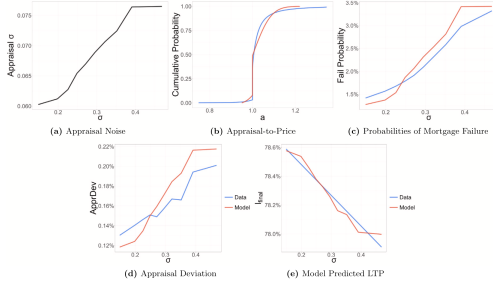


Fig. 7. Model Fit.
Note: Panel (a) shows estimated appraisal standard deviations σ_a on the y-axis, and estimated idiosyncratic price dispersion σ on the x-axis. Panel (b) shows the distribution of appraisal-to-price ratios a in the data and the fitted model, for the 5th σ -decile (that is, counties with values of σ between the 40th and 50th percentiles). Panel (c) shows transaction failure probabilities in the data and in the fitted model. Panel (d) shows $AppDev$, which is defined as $\frac{App - \sigma_a}{\sigma_a}$ in the data and in the fitted model. Panel (e) shows average values of l_{final} from the model, along with a line with slope equal to the l_{final} -to- σ regression coefficient we use as a target moment, normalized to have mean equal to the model-predicted loan-to-price ratio.

Table 7

Model estimates.

This table presents the estimates of model parameters. Panel A reports values of externally calibrated parameters. Panel B reports values of estimated parameters. The parentheses in Panel B show 95% confidence intervals, calculated from 200 bootstrapped moment samples, as we describe in Appendix H.2.1.

Panel A: Externally Calibrated Parameters

Description	Parameter	Value
Intertemporal elasticity of substitution	η	2
Wealth at time of home purchase	W_1	\$60,000
House price	P	\$200,000
Discount factor	β	0.96
	T	7
Maximum LTV parameter	ϕ	0.8

Panel B: Parameters Calibrated to the Data or through Moment Matching

Description	Parameter	Value
Appraisal Standard Deviation	$\sigma_1, \dots, \sigma_{10}$	See Fig. 7
Search cost	ζ	0.583 (0.306, 1.016)
Appraisal Bias	b	0.086 (0.086, 0.088)
Penalty rate on consumption	ψ	2.782 (2.022, 3.794)
Marginal utility of next period consumption	u'_2	0.00243 (0.00227, 0.00250)

8.5 Figure 7: model fit diagnostics

Fig. 7 checks that the calibrated model replicates key moments:

- (a) appraisal noise $\hat{\sigma}_a$ rises with price dispersion σ ;
- (b) the CDF of appraisal-to-price ratios is matched well (including bunching at $a = 1$);
- (c) failure probabilities increase with σ and the model matches their level;
- (d) appraisal deviation $AppDev$ rises with σ and the model tracks it;
- (e) average final leverage l_{final} declines with σ , matching the targeted regression slope.

9 Decomposing channels: how dispersion moves each outcome

9.1 Figure 8: decomposition logic

The paper uses counterfactual toggles:

- **Collateral recovery channel:** allow $r(l, \sigma)$ to vary with σ but *shut down appraisal risk* by holding appraisal noise constant across σ -deciles.
- **Appraisal risk channel:** hold the rate menu constant across σ but allow appraisal distributions to vary; then split the appraisal effect into: (i) *ex-ante* (borrowers choose lower target l when appraisal risk is high), and (ii) *ex-post* (realized under-appraisals force $l_{final} < l$).

9.2 Key findings from Figure 8

1. **Interest rates: mostly collateral recovery.** The σ -interest-rate relationship is driven primarily by the rate menu shifting upward when dispersion is higher.

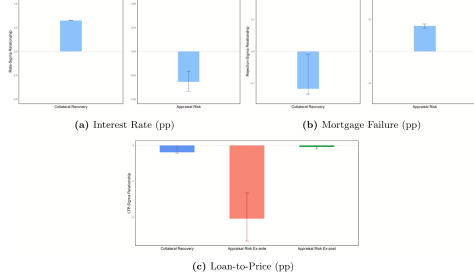


Fig. 8. Decomposition of Channels.
Notes: Panel (a), (b), and (c) respectively decompose the effect of price dispersion on interest rates, mortgage rejection rates, and loan-to-price ratios, into components attributable to the collateral recovery channel, and the appraisal channel. In all panels, the y-axis is the change in the outcome variable when sigma increases by 1. Note that the units differ from those in our reduced-form results, which standardize sigma to convert the results in this figure to be comparable to the reduced-form results, all numbers should be multiplied by the standard deviation of sigma, which is 0.11. To calculate the effect of the collateral recovery channel, we shut off the effect of the appraisal channel by setting the appraisal standard deviation parameter to a constant across σ -deciles, but allowing the interest rate menu to vary across σ -deciles. Analogously, to calculate the effect of the appraisal risk channel, we shut off the collateral recovery channel, by assuming the rate menu does not vary across σ -deciles, allowing only appraisal standard deviations to vary. In Panel (c), we further decompose the appraisal risk channel into an ex-ante effect, which measures how target loan size l^* varies with interest rate buckets, due to buyers' precautionary decisions to decrease loan size; and an ex-post effect, which measures how the $\text{pp}(l_{\text{final}} < l^*)$ changes with σ , which measures how realized under-appraised loan size. Note that, in both panels (a) and (b), the y-axes are not the same in the collateral recovery and appraisal risk panels.

Table 8

Automated appraisals counterfactuals.

This table shows how different versions of automated appraisals influence market outcomes, counterfactually within our calibrated model. In Panel A, we show results assuming that automated appraisals simply remove human biases from appraisals. In Panel B, we show results assuming that human biases are removed, the appraisal mean is shifted within each σ -decile such that the probability of underappraisal is unchanged, and the variance of appraisals is halved, so underappraisals are on average smaller when they occur. In each panel, the first two columns show the percentage point changes in final loan-to-price ratios l_{final} and interest rates, relative to the baseline outcome, and the third column shows the net increase in mortgage failure probability. The last column shows the “compensating variation” in prices, relative to the status quo of human appraisers, needed to make the consumer indifferent to the shift to automated appraisers. Quantities in parentheses reflect 95% confidence intervals, calculated from 200 bootstrapped moment samples, as we describe in Appendix H.2.1.

Panel A: Removing Bias				
	Loan-to-price Ratio (pp)	Interest Rate (pp)	Fail probability (pp)	Required compensating price change (%)
1st σ -bucket	-2.773 (-3.044, -2.367)	-0.052 (-0.057, -0.043)	10.54 (9.188, 12.398)	-5.347 (-6.875, -3.914)
5th σ -bucket	-2.616 (-2.827, -2.226)	-0.049 (-0.053, -0.04)	11.964 (10.633, 13.905)	-5.786 (-7.463, -4.216)
10th σ -bucket	-2.36 (-2.533, -2.035)	-0.044 (-0.048, -0.037)	13.561 (12.145, 15.44)	-6.262 (-8.126, -4.536)

Panel B: Reducing Variance				
	Loan-to-price Ratio (pp)	Interest Rate (pp)	Fail probability (pp)	Required compensating price change (%)
1st σ -bucket	0.166 (0.062, 0.241)	0.003 (0.001, 0.005)	-0.713 (-0.752, -0.66)	0.364 (0.201, 0.595)
5th σ -bucket	0.105 (0.052, 0.229)	0.002 (0.001, 0.004)	-1.079 (-1.119, -0.997)	0.5 (0.278, 0.81)
10th σ -bucket	0.109 (0.023, 0.197)	0.002 (0, 0.004)	-1.6 (-1.684, -1.496)	0.694 (0.39, 1.086)

2. **Failures and leverage: mostly appraisal risk.** Mortgage failures (transaction failures/rejections) and loan-to-price declines are dominated by appraisal risk, especially the *ex-ante* component: buyers in high-dispersion areas choose lower leverage in anticipation of potential under-appraisals.
3. **Ex-post appraisal pressure is small.** The ex-post wedge (realized appraisal forcing $l_{\text{final}} < l^*$) is quantitatively small relative to the ex-ante leverage adjustment.

10 Policy counterfactuals and welfare interpretation

10.1 Compensating variation: turning utility costs into price changes

The paper uses a “compensating variation” calculation: for a buyer in a given σ -decile, compute how much the house price would have to fall to keep expected utility constant relative to a low-dispersion benchmark.

10.1.1 Figure 9: compensating variation increases in dispersion

Fig. 9 plots the required percentage price decrease across σ -deciles. The required price drop becomes larger (more negative) in higher-dispersion buckets, implying sizable utility costs of collateral value uncertainty.

10.2 Automated appraisals (Table 8): removing bias vs reducing variance

Table 8 uses the calibrated model to evaluate two stylized versions of automated appraisal systems.

10.2.1 Panel A: removing human bias ($b = 0$)

If automated appraisals eliminate upward bias and bunching, then under-appraisals become more frequent and more severe. The model predicts:

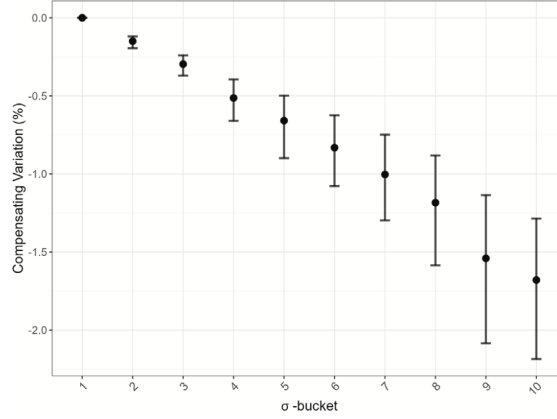


Fig. 9. Price Dispersion and Compensating Variation in Prices.

Notes: This figure presents the “compensating variation” in prices needed to offset the costs of price dispersion; that is, in each σ -decile, we show how much prices would have to decrease, in order for consumers to attain the same expected utility as in the first σ -decile. Error bars show 95% confidence intervals, calculated from 200 bootstrap moment samples, as we describe in Appendix H.2.1.

- large declines in loan-to-price (about -2.4 to -2.8 pp),
- large increases in failure probability (about $+10$ to $+14$ pp),
- slightly lower interest rates (because borrowers respond by choosing smaller loans),
- large compensating price decreases needed to offset the welfare loss (around -5% to -6%).

10.2.2 Panel B: reducing appraisal variance

If automated appraisals reduce appraisal noise (variance) while keeping the mean and under-appraisal probability roughly stable, then outcomes improve modestly:

- small increases in loan-to-price (around $+0.10$ to $+0.17$ pp),
- small decreases in failure probabilities (around -0.7 to -1.6 pp),
- small compensating *increases* in prices buyers would be willing to pay (about $+0.36\%$ to $+0.69\%$).

Policy intuition. Bias and bunching in human appraisals are not “pure inefficiency” in this model: they partially insure borrowers against small under-appraisals that would otherwise bind regulatory LTV constraints. Therefore, automation that removes bias without reducing variance can worsen credit outcomes; automation that truly reduces variance can improve them.

11 Conclusion: what to remember for future use

Empirical takeaway. Collateral value uncertainty—measured as idiosyncratic house price dispersion—tightens mortgage credit: higher rejection rates, higher rates conditional on leverage, and lower leverage.

Identification takeaway. A “house non-standardization” instrument can shift dispersion within county-year without shifting borrower credit quality, supporting a causal interpretation.

Mechanism takeaway. Two channels matter, but for different outcomes:

- Interest rate effects are mainly *collateral recovery*.
- Rejections and leverage effects are mainly *appraisal risk* interacting with regulatory constraints.

Policy takeaway. Appraisal reforms must be evaluated through both mean/bias and variance/noise: removing human bias can backfire if it increases under-appraisal pressure; reducing noise is beneficial.

12 Conclusion

12.1 Summary

This paper concludes that **collateral value uncertainty** is an economically important and distinct determinant of mortgage credit provision. Even holding constant borrower characteristics, lender fixed effects, and local macro conditions, mortgages collateralized by houses whose values are *harder to pin down* receive **less credit, at worse terms, and with higher failure risk**.

The paper organizes this conclusion around a single causal chain:

1. **Measure collateral value uncertainty at the property level.** Using a flexible hedonic pricing model, the paper constructs a house-level (and then zipcode-year) measure of *idiosyncratic house price dispersion*—the predicted conditional variance of transaction-price residuals. This dispersion is persistent and systematically higher in disadvantaged areas, implying that collateral uncertainty is not random in the cross-section.
2. **Show that dispersion increases appraisal noise and under-appraisal risk.** Higher dispersion is associated with larger appraisal deviations (both under- and over-appraisals), with the downside tail (under-appraisals) being crucial for credit constraints.
3. **Establish reduced-form causal effects on credit provision.** Using mortgage applications and originations, the paper shows that a higher-dispersion zipcode (or house) implies:
 - **Higher rejection rates**, especially *collateral-related* rejections;
 - **Higher interest rates conditional on leverage** (an upward shift in the rate-LTP menu);
 - **Lower loan-to-price ratios** (lower leverage).

An IV strategy based on *house non-standardization* (distance of house features from county-year norms) strengthens these effects and passes an exclusion test using ex-post default: in IV, dispersion does not predict default, supporting an interpretation as *collateral uncertainty* rather than borrower risk.

4. **Quantify mechanisms with a structural model.** A calibrated model that mirrors the mortgage-origination pipeline (choice of target leverage $\rightarrow$ rate menu $\rightarrow$ appraisal realization $\rightarrow$ LTV constraint using $\min\{P, A\} \rightarrow$ continuation/renegeing) rationalizes the empirical patterns with two mechanisms:

- (a) a **collateral recovery channel** (lenders price more uncertain collateral more conservatively, shifting interest-rate menus upward), and
- (b) an **appraisal risk channel** (noisy appraisals interacting with LTV constraints raise failure risk and induce borrowers to choose lower leverage *ex ante*).

The model implies that **interest-rate effects are mostly driven by collateral recovery**, while **rejections and leverage declines are mostly driven by appraisal risk**, especially the *ex-ante* leverage adjustment.

5. **Welfare and policy.** Collateral uncertainty generates meaningful utility losses (a compensating-variation mapping implies buyers would require sizable effective price discounts in high-dispersion markets). Moreover, appraisal reforms must be evaluated jointly in terms of *bias* and *noise*: removing human appraisal bias/bunching without reducing variance can *worsen* credit outcomes by increasing under-appraisal pressure, whereas reducing appraisal variance improves outcomes (typically more modestly).

12.2 Economic intuition: why “uncertain collateral” reduces mortgage credit

The paper’s intuition is that mortgages are constrained not only by *borrower risk* but also by *collateral measurement and enforceability*.

1) Collateral recovery: uncertainty raises the price of leverage. If the collateral value is more uncertain, then the lender faces more downside risk (or lower expected recovery) in foreclosure. Competitive lending implies a higher required rate at a given leverage:

$$r(l, \sigma) \uparrow \text{ in } \sigma,$$

so the entire interest-rate menu shifts upward when dispersion σ is higher.

2) Appraisal risk: under-appraisals make leverage risky even for good borrowers. Regulatory and securitization constraints effectively apply to

$$\text{LTV computed on } \min\{P, A\},$$

so a low appraisal ($A < P$) mechanically reduces the maximum feasible loan amount. This creates a **down-payment shock**:

$$\text{extra down payment} \propto \max\{0, l - \phi(A/P)\},$$

which is especially painful because it is *unplanned* (liquidity shock), and may force the deal to fail if the borrower cannot bridge the gap.

3) Why borrowers choose lower leverage ex ante. Rational borrowers anticipate that in high-dispersion markets, appraisals are noisier and the probability of binding under-appraisals is higher. Thus they optimally reduce target leverage l to reduce the chance of an appraisal-induced constraint or failure:

$$l^*(\sigma) \downarrow \text{ in } \sigma.$$

This is why the model attributes most of the leverage decline to an *ex-ante* channel rather than an ex-post mechanical adjustment.

4) Why “bunching” at the transaction price can be privately beneficial. Empirically, appraisals bunch at $A = P$. In the model, this acts like partial insurance against small under-appraisals that would otherwise bind LTV rules. Hence, eliminating this feature without reducing appraisal noise can *increase* under-appraisal events and worsen credit provision.

12.3 What to remember

- **Conceptual:** mortgages are a setting where the *measurement* of collateral (appraisals) matters as much as its level.
- **Empirical:** higher collateral dispersion causes higher rejection (especially collateral-related), higher rates at given leverage, and lower leverage.
- **Mechanisms:** rates $\Rightarrow$ mostly collateral recovery; failures and leverage $\Rightarrow$ mostly appraisal risk $\times$ LTV constraints.
- **Distributional:** because dispersion is higher in disadvantaged neighborhoods, collateral uncertainty can propagate inequality in credit access.
- **Policy:** appraisal reform that reduces *noise* is beneficial; removing *bias/bunching* without noise reduction can backfire.

12.4 Takeaway

Jiang and Zhang (2025) show that idiosyncratic collateral value uncertainty—captured by house price dispersion—tightens mortgage credit by raising collateral-related rejections, shifting up interest-rate menus, and lowering leverage, with structural evidence that appraisal risk interacting with LTV rules is the dominant driver of failures and leverage reductions, while collateral recovery risk mainly explains interest-rate effects, implying appraisal-noise reduction (not simply bias removal) is the key margin for improving credit provision.